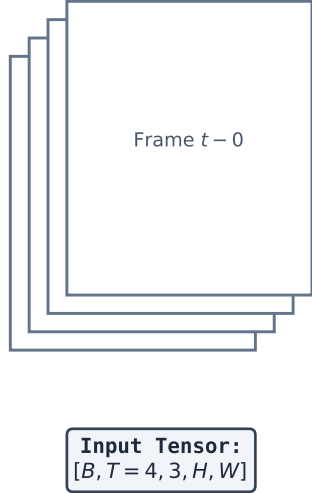
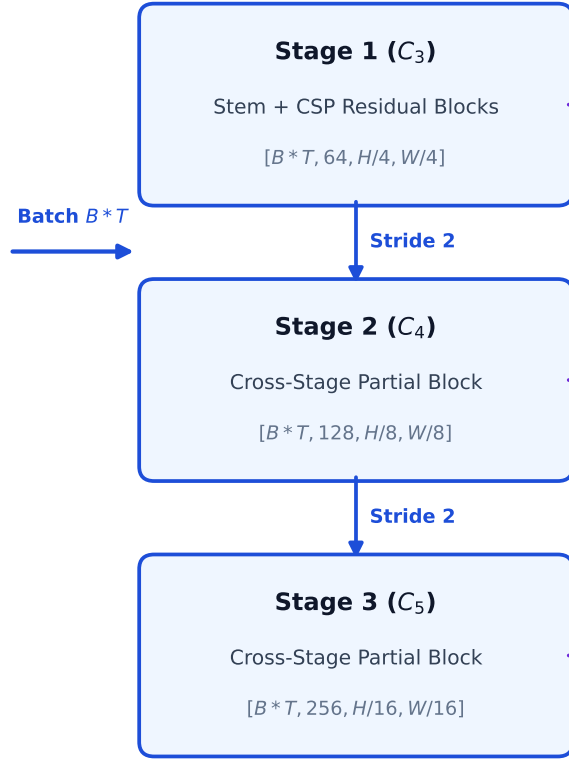


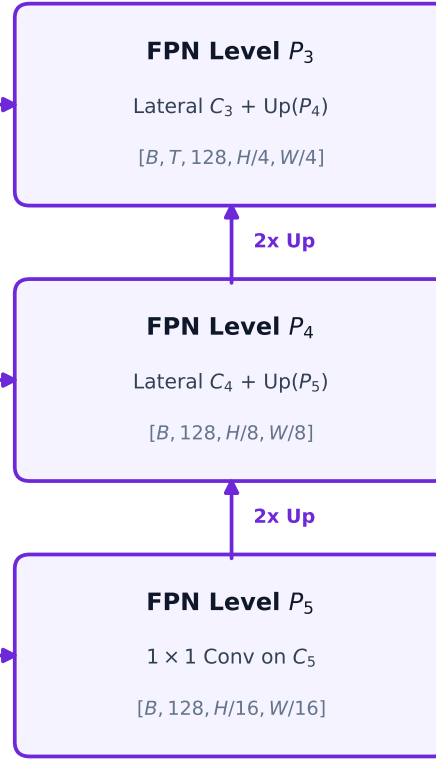
(a) Input Sequence



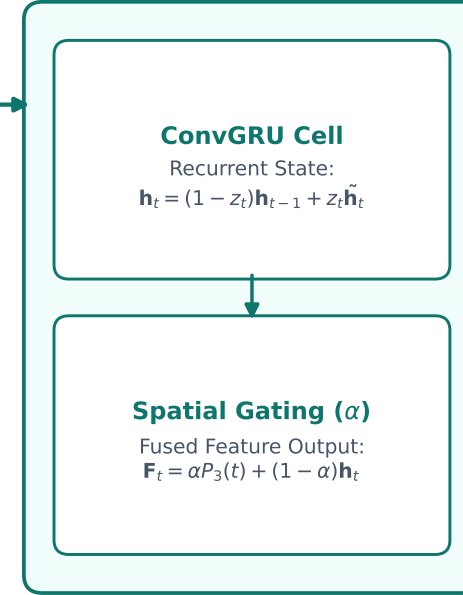
(b) CSP-Darknet Backbone



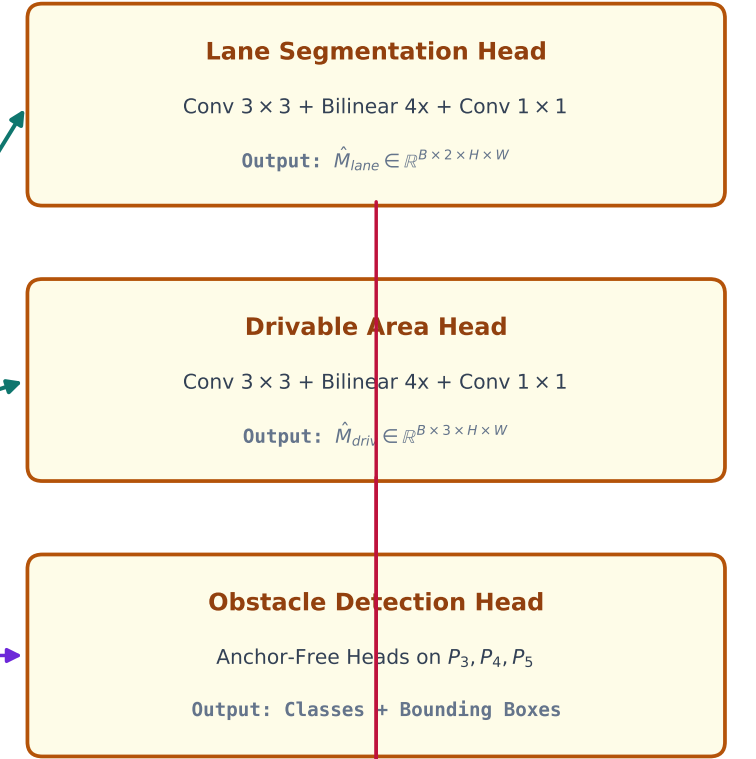
(c) Feature Pyramid (FPN)



(d) ConvGRU Aggregator



(e) Multi-Task Decoders

Cross-Task Consistency Regularization Loss ($\mathcal{L}_{CTC}$)

$$\mathcal{L}_{CTC} = \frac{1}{|P|} \sum_p \sigma(\hat{M}_{lane}(p)) \cdot [1 - \text{Dilate}(\sigma(\hat{M}_{drivable}(p)), k)]$$

(Prevents lane predictions outside drivable road boundaries)